Date: 2026-10-12
To: Editor-in-Chief, IEEE Transactions on Information Forensics and Security
Manuscript ID: T-IFS-28362-2026 (existing draft; not a new submission)
Title: Probing-Resistant Masked Hardware: Formal Security Proofs and Simulation-Based Side-Channel Verification

Dear Editor,

Following the editorial notice of 9 October 2026, the appendix previously uploaded as a separate file has been relabeled as *Supplemental Material* and is uploaded as such. The main manuscript is self-contained and is at most 13 pages in the two-column IEEE format.

Contributions.

- A characterization of the soundness gap between equipment-free value-based statistical verification and glitch-extended probing, via a cone-non-completeness condition; a first-order lint shown sound, and an explicit first-order witness showing that the gap is real.
- A masked multiplication gadget (DOM with register barrier) proven robust d-NI with perfect simulation for arbitrary d, and a first-order AES S-box built from it.
- A reproducible, equipment-free verification flow for the first-order design combining the lint, PROLEAD and gate-level TVLA/MI, calibrated with an unmasked positive control.

Scope. The empirical evaluation targets the first-order design; the gadget-level proof holds for arbitrary d, and the lint theorem is stated at first order. Limitations are discussed in the paper.

Supplemental Material. Corrected PROLEAD worst-case value, the explicit gap witness, the pre-barrier PROLEAD history, extended discussion, further verifiers and full-chip details.

Artifact. RTL, synthesis scripts, lint, PROLEAD configurations and results are available at `https://github.com/mipsan1/probing-resistant-masked-hardware` and in the artifact bundle accompanying the submission.

The manuscript is not published elsewhere or under consideration at another journal. The author approves this submission. No conflict of interest.

Sincerely,
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